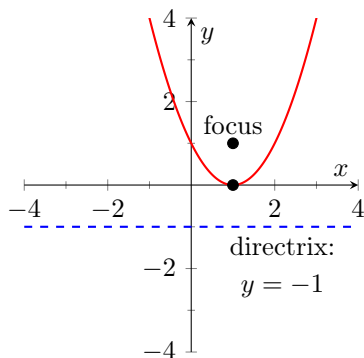


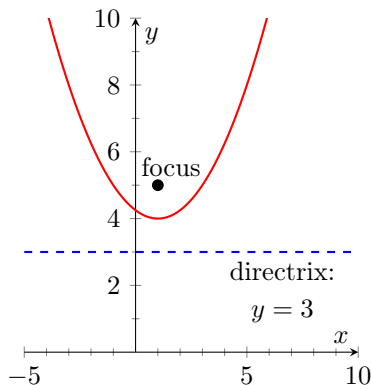
1. Find and graph the equation of the parabola with focus $(1, 1)$ and directrix $y = -1$.



$y = \frac{1}{4p}(x - h)^2 + k$ where the vertex is (h, k) and the distance from the vertex to the focus is p .

- The distance from the focus to the directrix is $1 - (-1) = 2$ so $p = \frac{1 - (-1)}{2} = 1 \Rightarrow a = \frac{1}{4}$.
- $k = 1 - p = 0$ (the y -coordinate of the focus minus p)
- $h = 1$ (the x -coordinate of the focus)
- So, $y = \frac{1}{4}(x - 1)^2$

2. Find the focus and directrix of the parabola $y = \frac{1}{4}(x - 1)^2 + 4$.



$a = \frac{1}{4} \Rightarrow \frac{1}{4} = \frac{1}{4p} \Rightarrow p = 1$, and the vertex is $(1, 4)$.

- The focus is $(h, k + p)$: $(1, 5)$
- The directrix is $y = k - p$: $y = 3$